

MEDICINE LODGE, KS, USA

WMO: 724520

Lat: 37.267N

Lon: 98.548W

Elev: 468

StdP: 95.83

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03957

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-10.7	-19.0	0.7	-10.3	-16.8	0.9	-8.4	12.9	3.6	11.6	3.9	3.4	0	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	39.1	22.3	37.4	22.4	35.7	22.4	24.8	33.0	24.1	32.6	23.5	31.9	5.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.3	27.9	21.8	17.5	27.1	21.2	16.8	26.8	77.7	33.1	74.8	32.5	72.4	31.8	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.9	10.6	9.1	-17.3	41.4	3.3	1.8	-19.6	42.7	-21.5	43.7	-23.4	44.7	-25.8	46.0		
			-17.7	26.0	3.1	0.7	-20.0	26.5	-21.8	26.9	-23.6	27.3	-25.8	27.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	1.4	3.0	8.7	14.1	19.6	25.3	27.9	27.0	22.2	14.9	8.0	2.0
	DBStd	10.65	5.64	6.42	5.95	5.14	4.76	3.43	3.24	3.28	4.42	5.07	5.28	5.30
	HDD10.0	953	268	208	93	19	1	0	0	0	1	14	97	251
	HDD18.3	2401	524	430	301	144	42	1	0	0	15	128	310	506
	CDD10.0	2624	3	12	54	142	300	458	554	526	367	168	38	3
	CDD18.3	1030	0	0	3	17	83	209	296	267	132	23	1	0
	CDH23.3	12275	1	6	51	240	899	2419	3782	3173	1402	282	21	0
	CDH26.7	6364	0	1	9	74	377	1249	2181	1738	642	90	2	0
Wind	WSAvg	4.1	3.9	4.3	4.7	5.1	4.4	4.3	3.7	3.5	3.9	4.0	4.0	3.9
Precipitation	PrecAvg	664	15	23	51	63	90	102	72	84	63	55	42	22
	PrecMax	915	52	75	231	181	235	208	211	209	190	184	154	101
	PrecMin	346	0	0	0	6	6	16	0	10	4	0	0	0
	PrecStd	129	14	22	47	44	59	50	49	61	46	45	40	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.8	23.9	27.9	32.5	36.4	39.6	41.4	40.8	38.1	32.3	25.9	19.4
		MCWB	9.8	11.8	14.4	16.8	20.1	22.0	22.3	22.3	21.3	18.3	14.2	10.7
	2%	DB	16.9	20.2	24.5	28.6	33.4	36.6	39.5	38.5	34.9	29.2	22.9	15.9
		MCWB	8.1	10.6	14.3	16.2	20.3	22.2	22.4	22.3	21.3	17.8	13.5	8.8
	5%	DB	13.8	16.7	21.8	26.1	30.9	35.0	37.8	36.8	32.8	26.6	19.9	13.3
		MCWB	6.6	9.2	13.1	16.0	20.0	22.2	22.7	22.5	21.1	16.8	12.1	7.7
	10%	DB	10.8	13.4	18.9	23.4	28.4	33.2	35.9	34.8	30.5	23.9	17.1	10.7
		MCWB	5.2	7.3	11.9	14.9	19.3	21.9	22.8	22.5	20.2	16.2	11.3	6.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	14.6	18.4	20.2	23.1	25.2	25.8	25.4	23.9	21.8	16.8	13.7
		MCDB	17.5	19.2	24.2	26.3	30.6	33.4	33.6	33.3	31.9	26.8	21.2	15.9
	2%	WB	8.8	11.8	16.6	19.0	22.2	24.1	25.0	24.6	22.9	20.2	15.2	10.1
		MCDB	15.8	17.6	20.6	24.6	29.5	32.3	33.1	33.0	30.4	25.2	19.2	14.3
	5%	WB	7.1	10.0	14.8	17.7	21.3	23.5	24.1	23.9	22.2	18.8	13.7	8.0
		MCDB	13.2	17.1	19.4	23.4	28.2	31.5	32.8	32.4	29.5	23.6	18.3	12.7
	10%	WB	5.5	7.6	12.5	16.3	20.4	22.7	23.5	23.3	21.4	17.2	11.9	6.2
		MCDB	10.5	13.2	18.4	22.1	26.5	30.6	32.4	31.3	28.4	22.5	16.8	10.6
Mean Daily Temperature Range	5% DB	MDBR	13.1	13.2	13.5	13.6	12.9	13.1	13.2	13.0	13.3	13.8	13.7	12.2
		MCDBR	18.3	18.8	18.0	17.5	15.7	15.3	15.9	15.8	15.2	17.4	17.9	16.6
		MCWBR	10.9	10.9	9.8	8.6	6.3	4.6	4.0	4.4	5.0	8.1	9.7	10.5
	5% WB	MCDBR	16.9	17.5	14.2	14.2	13.3	13.2	13.3	13.0	12.9	13.0	14.4	15.1
		MCWBR	10.5	10.7	8.9	8.3	6.5	4.9	4.5	4.6	4.8	7.3	8.8	10.2
Clear-Sky Solar Irradiance	taub	0.281	0.291	0.333	0.353	0.378	0.403	0.410	0.405	0.370	0.340	0.299	0.278	
	taud	2.514	2.496	2.394	2.376	2.389	2.339	2.347	2.315	2.399	2.460	2.508	2.528	
	Ebn at Noon	920	951	932	927	904	877	868	865	880	879	887	900	
	Edh at Noon	78	90	110	119	120	126	124	125	109	93	78	72	
All-Sky Solar Radiation	RadAvg	2.65	3.39	4.48	5.58	6.31	7.04	6.92	6.19	5.15	3.82	2.87	2.31	
	RadStd	0.25	0.45	0.33	0.34	0.53	0.49	0.39	0.35	0.33	0.44	0.32	0.22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page